

ANIMAL MODELS VERIFY THE GENETIC BASIS FOR CUTANEOUS NEOPLASMS

Using human hereditary cancer syndromes as a guide, genetically based animal models have been developed that validate the contribution of individual genes to human cancer susceptibility by prospective experimental analysis. These models have also contributed to the understanding of tumor initiation, progression, and the role of specific genetic alterations in the context of cutaneous neoplasms.

Gene Targets for Mutations in Hereditary and Sporadic Cutaneous Cancers

GENE	FUNCTION	LOCUS	TUMOR TYPE	SYNDROME	SPONTANEOUS
P53	DNA repair, apoptosis, cell cycle regulation	17p13.1	BCC, SCC	Li-Fraumeni (but no increase in skin cancers)	Yes
	XPA, XPB	DNA repair	9q22.3, 2q21	BCC, SCC, melanoma	Xeroderma pigmentosum
	XPC, XPD	DNA repair	3p25, 19q13.2-q13.33	BCC, SCC, melanoma	Xeroderma pigmentosum
	XPF, XPG	DNA repair	16p13.3-p13.13, 13q33	BCC, SCC, melanoma	Xeroderma pigmentosum

The importance of DNA as a target for carcinogenesis is strongly supported by the discovery of defects in DNA repair genes in xeroderma pigmentosum (XP) families. At least six distinct genes—classified into complementation groups—encode proteins essential for nucleotide excision repair (NER), a critical pathway protecting DNA from damage induced by ultraviolet radiation (UVR) and chemical carcinogens. These include:

- **Damage recognition proteins:** XPA, XPC, XPE
- **Helicases:** XPB, XPD
- **Endonuclease components:** XPG, XPF

Defects in any of these genes result in a skin-cancer-prone phenotype. Functional polymorphisms in these and other DNA repair genes may also influence cancer susceptibility in the general population.

Analysis of squamous cell carcinomas (SCCs) and basal cell carcinomas (BCCs) from XP patients has revealed frequent signature mutations in key genes such as **PTCH1**, **RAS**, **p53**, and **INK4a-ARF**, highlighting their roles as potential targets in UV-induced carcinogenesis.

Genetic studies in basal cell nevus syndrome (Gorlin syndrome), along with research in *Drosophila* and mice, have implicated **PTCH1** mutations and dysregulation of the **Hedgehog (Hh) signaling pathway** in both hereditary and sporadic BCCs, as well as certain internal malignancies. These tumors consistently show elevated Hh signaling activity.

The **INK4a-ARF** locus, particularly **INK4A** mutations, has been linked to melanoma pathogenesis through familial melanoma inheritance studies.

Additional hereditary syndromes associated with adnexal and other skin tumors include:

- **Cowden syndrome:** *PTEN* mutations
- **Muir-Torre syndrome:** *MSH2*, *MLH1* mutations
- **Multiple cylindromatosis:** *CYLD1* mutations
- **Multiple trichoepithelioma:** *CYLD1* mutations
- **Birt-Hogg-Dubé syndrome:** *FLCN* mutations
- **Sporadic pilomatricomas:** *CTNNB1* (β -catenin) mutations
- **Sebaceous tumors:** *LEF1* involvement

Chromosomal mapping in these and other syndromes (see Table 110-2) continues to clarify the molecular basis of skin neoplasia, offering insights into both inherited and sporadic forms of disease.